

SECOND ROBOTICS PROJECT

ROBOTICS



POLITECNICO
MILANO 1863

THE PROBLEM



Provided data:

- Odometry from the robot
- tf
- Laser scan data



DATA



Format: ROS Bag file

bag available here:

<https://drive.google.com/file/d/1Ik1HoX3EpYHnnP8jxwnrkqgFbleDM1ss/view?usp=sharing>

Data:

- /odom: odometry data from the robot (noisy)
- /ugv/rslidar_points: pointcloud from lidar
- /tf: dynamic tf
- /tf_static: static tf, like sensors position



THE PROJECT (Task 1: mapping)

- Use the bag to create a map of the environment
 - Use the preferred mapping package
 - Write a launch file that starts:
 - all required nodes to perform data conversion
 - the mapping node
 - rviz with config file to show the map, the lidar and the tf, set global frame to map



THE PROJECT (Task 1: mapping)

- To build a map you need tf and laserscan:
 - The bag contains a pointcloud topic
 - you can use a node, e.g., pointcloud2laserscan to convert the data to laserscan



THE PROJECT (Task 2: navigation)

- Setup a **realistic** simulation of the robot using stage
- Robot size: AgileX Scout Mini Omnidirectional
- Robot kinematics: Omnidirectional
- Setup the navigation stack to receive goals and move the simulated robot avoiding obstacles in the generated map
- Write a goal-publisher node that reads a sequence of goals from a csv and send them to the robot. A new goal is sent when the robot reach the previous one or it's aborted
- csv file will be in the folder: `second_project/csv/goals.csv`
- an example csv file is provided



THE PROJECT (Task 2: navigation)

- Provide a launch file that starts:
 - stage simulation with the robot and the map you build during task 1
 - nav2 configured to localize in the provided map and drive autonomously the robot avoiding obstacles
 - the controller node that publish the goal after reading them from csv, **using action**
 - csv structure: x,y,theta
 - rviz configured to visualize the map, the tf's, the particle cloud (if amcl is used), the laser scanner, the paths and the goals



THE PROJECT (Task 2: navigation)

- Also provide a map folder with:
 - png file of the reconstructed map (**mandatory**)
 - serialized map if slam toolbox is used



Deadlines and requested files

- Upload **only** a tar.gz file to webeep (**only one team member upload the files**)
- Inside the archive:
 - info.txt file (details next slide)
 - folders of the nodes you created (with inside CmakeLists.txt, package.xml, etc...)
 - map folder
 - **do not upload** the entire environment (with build and devel folders)
 - **do not upload** the bag files



Deadlines and requested files

File txt must contain only the group names with this structure

codice persona;name;surname

You can add another file called readme.txt with additional info. I will not always look for it. But if something goes wrong I'll check for explanations.



Some more requests

Name the archive with your **codice persona**

Name the package **second_project**

Don't use absolute path

The project need to be written using c/c++



Deadlines and requested files

Deadline: 18 June (4 weeks)

Max 3 student for team

N.B.: If the grading is needed earlier you can submit the project before the deadline. Then write us a mail and specify the need for earlier grading in the message and mail title

Questions:

- write to me via mail (simone.mentasti@polimi.it)
- do not write only to Prof. Matteucci



How to get less points

- The project do not compile -> 0 points
- The project has absolute paths -> 0 points
- The archive has build and devel folder -> -1 points
- The archive has bag files -> -1 points
- The project do not open rviz -> - 1 points
- The project do open rviz with wrong config -> - 1 points
- Two members upload the project -> $\frac{1}{2}$ points
- Three members upload the project -> $\frac{1}{3}$ points